Magnets

Exam code: 0625 & 0972

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Updated on 10 February 2026

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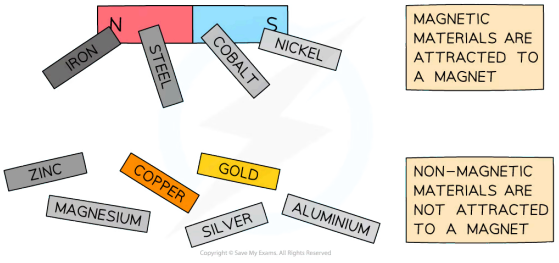
Induced magnetism

Magnetic materials

- Very few metals in the Periodic Table are magnetic
- The magnetic metals are:
  - Iron
  - Cobalt
  - Nickel
  - Steel is an alloy which contains iron, so it is also magnetic

Magnetic materials



Magnetic materials are attracted to a magnet; non-magnetic materials are not

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Got it

Magnets and magnetic materials

Magnetic materials are attracted to both poles of a permanent magnet

- Bringing a material close to a known magnet will determine if the material is magnetic, non-magnetic or if it is a magnet itself
  - It is a **magnet** if it is **repelled** by the known magnet
  - It is a **magnetic material** if it is **attracted** to the known magnet
  - It is a **non-magnetic material** if it is **neither** attracted nor repelled by the known magnet

Types of magnets

- There are two types of magnets:
  - Permanent magnets
  - Induced (also called temporary) magnets

Permanent magnets

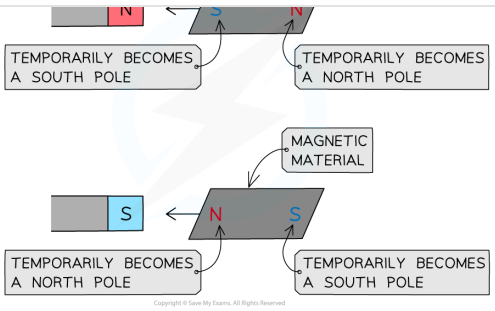
- Permanent magnets are made out of permanent magnetic materials, for example, **steel**
- A permanent magnet will **produce its own magnetic field**
  - It will not lose its magnetism

Temporary (induced) magnetism

- An **induced magnet** is a material with a **soft iron core** that **becomes** a magnet **temporarily** when it is placed in a magnetic **field**
  - Induced magnetism always causes a force of **attraction** between the **permanent magnet** creating the magnetic field and the induced magnet
  - When this happens it is said that the material has been **magnetised**
  - This means that the end of the material closest to the magnet will have the **opposite** pole to that of the magnet pole closest to the material

- When removed from the magnetic field, the material will **lose** its induced magnetism **quickly** and become **unmagnetised**
  - Some objects such as paperclips or needles (made from steel) can be magnetised and will remain magnetic for a while
  - Other objects, such as electromagnets or transformers (which are made from **soft iron**) will be **unmagnetised** (no longer magnetised) as soon as the cause of the induced magnetism is **removed**

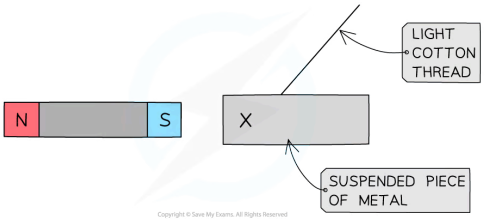
Induced magnetism in a magnetic material



Inducing magnetism in a magnetic material

Worked Example

The diagram below shows a magnet held close to a piece of metal suspended by a light cotton thread. The piece of metal is attracted towards the magnet.



Which of the following rows in the table gives the correct type of pole at X and the correct material of the suspended piece of metal?

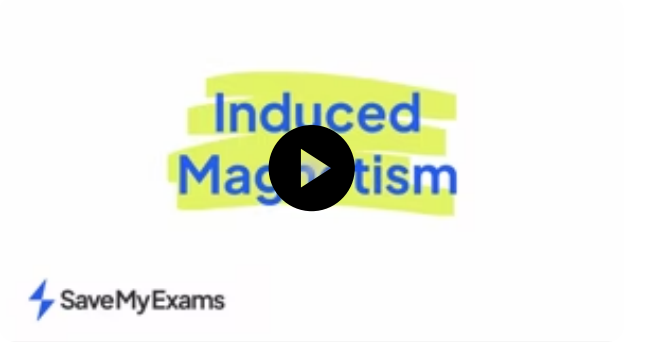
	Type of pole at X	Material of suspended piece of metal
A	North	Nickel
B	South	Nickel
C	North	Aluminium
D	South	Aluminium

Answer: A

- X must be a north pole
  - The piece of metal is being attracted towards the magnet
  - The law of magnetism states that opposite poles attract
- The material of the suspended piece of metal is nickel
  - Nickel is a magnetic material (It will experience a force when it is placed in a magnetic field, in this case it is attracted towards the magnet)

metal would be repelled from the magnet because the law of magnetism states that like poles repel

- **C** and **D** are incorrect because aluminium is not a magnetic material
  - A non-magnetic material would be unaffected by the magnetic field produced by the magnet



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